

GuestLecture_ANOVA_Markdown

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Data Storytelling: The Fundamentals of Data Visualization with ggplot2

We will be exploring the education data set that was retrieved from kaggle.com

(<https://www.kaggle.com/datasets/desalegngeb/students-exam-scores?resource=download>

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1. Load Packages

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(visdat)
library(corrplot)
```

```
## corrplot 0.92 loaded
```

2. Read in Data

```
education <- read.csv("education.csv")
head(education)
```

```
## student_ID Gender EthnicGroup ParentEduc LunchType TestPrep
## 1 0 female <NA> bachelor's degree standard none
## 2 1 female group C some college standard <NA>
## 3 2 female group B master's degree standard none
## 4 3 male group A associate's degree free/reduced none
## 5 4 male group C some college standard none
## 6 5 female group B associate's degree standard none
## ParentMaritalStatus PracticeSport IsFirstChild NrSiblings TransportMeans
## 1 married regularly yes 3 school_bus
## 2 married sometimes yes 0 <NA>
## 3 single sometimes yes 4 school_bus
## 4 married never no 1 <NA>
## 5 married sometimes yes 0 school_bus
## 6 married regularly yes 1 school_bus
## WklyStudyHours MathScore ReadingScore WritingScore School
## 1 < 5 71 71 74 1
## 2 5 to 10 69 90 88 1
## 3 < 5 87 93 91 1
## 4 5 to 10 45 56 42 1
## 5 5 to 10 76 78 75 1
## 6 5 to 10 73 84 79 1
```

3. Overview of Data

```
nrow(education)
```

```
## [1] 30641
```

```
ncol(education)
```

```
## [1] 16
```

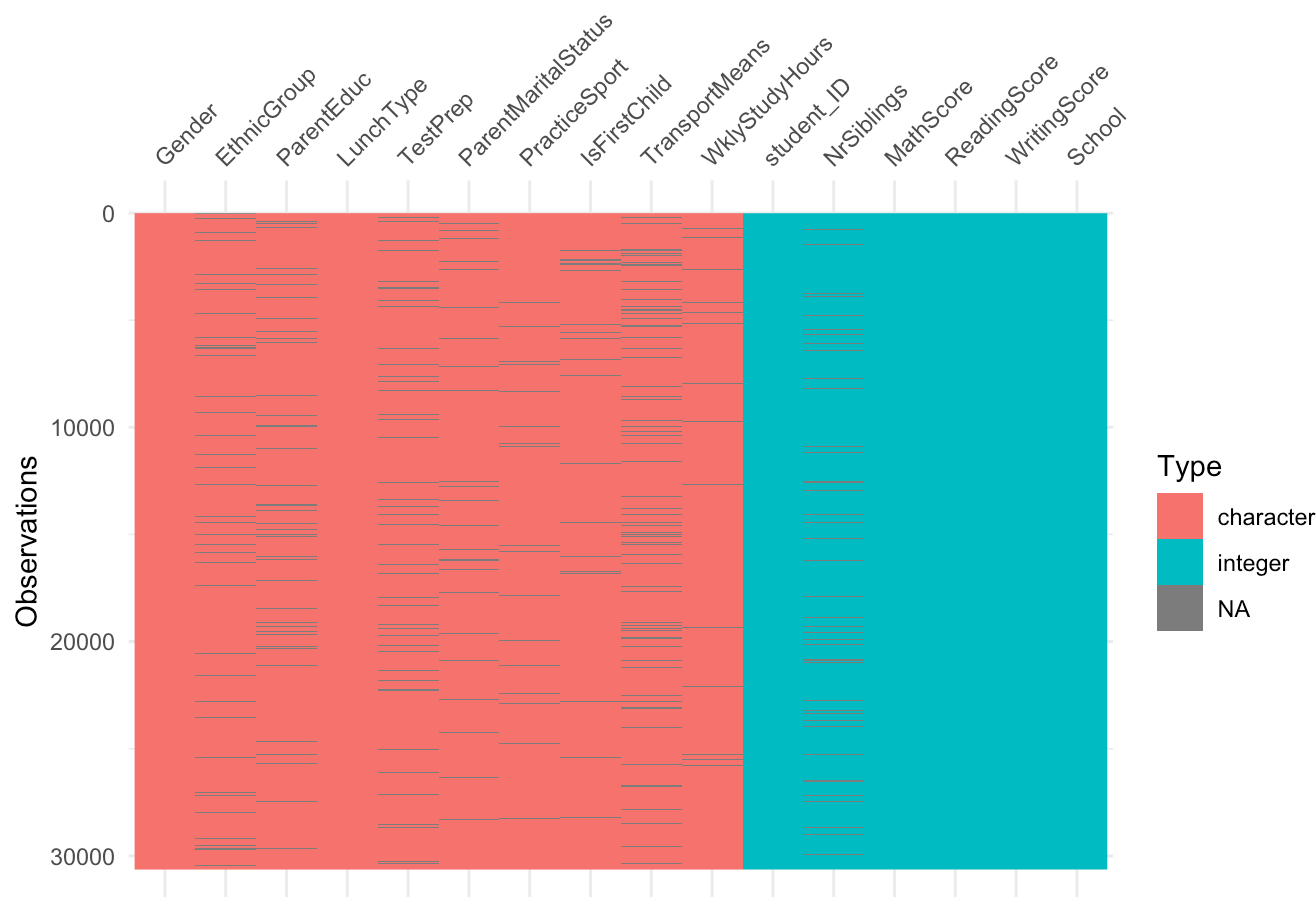
```
education %>% map_dbl(~sum( is.na(.) ) ) %>% sum()
```

```
## [1] 13901
```

30641 rows, 16 columns, and 13901 missing values.

4. Missingness

```
vis_dat(education)
```



5. Data Dictionary

- Gender: Gender of the student (2 categories: male or female)
- EthnicGroup: Ethnic group of the student (5 categories: group A, group B, group C, group D, group E)
- ParentEduc: Parent(s) education background (6 categories: some high school, high school, some college, associate's degree, bachelor's degree, master's degree)
- LunchType: School lunch type (2 categories: standard or free/reduced)
- TestPrep: Test preparation course followed (2 categories: none or completed)
- ParentMaritalStatus: Parent(s) marital status (4 categories: single, married, widowed, divorced)
- PracticeSport: How often the student practice sport (3 categories: sometimes, regularly, never)
- IsFirstChild: If the child is first child in the family or not (2 categories: yes or no)
- NrSiblings: Number of siblings the student has (8 categories: 0, 1, 2, 3, 4, 5, 6, 7)
- TransportMeans: Means of transport to school (2 categories: school bus or private)
- WklyStudyHours: Weekly self-study hours (3 categories: less that 5 hours, between 5 and 10 hours, or more than 10 hours)
- MathScore: Math test score (0-100)
- ReadingScore: Reading test score (0-100)
- WritingScore: Writing test score (0-100)
- School: 31 schools (31 categories: 1-31)

6. Change Datatypes

```
education %>% mutate(student_ID = as.factor(student_ID),
                      Gender = as.factor(Gender),
                      EthnicGroup = as.factor(EthnicGroup),
                      ParentEduc = as.factor(ParentEduc),
                      LunchType = as.factor(LunchType),
                      TestPrep = as.factor(TestPrep),
                      ParentMaritalStatus = as.factor(ParentMaritalStatus),
                      PracticeSport = as.factor(PracticeSport),
                      IsFirstChild = as.factor(IsFirstChild),
                      NrSiblings = as.factor(NrSiblings),
                      TransportMeans = as.factor(TransportMeans),
                      WklyStudyHours = as.factor(WklyStudyHours),
                      MathScore = as.numeric(MathScore),
                      ReadingScore = as.numeric(ReadingScore),
                      WritingScore = as.numeric(WritingScore),
                      School = as.factor(School)) -> education

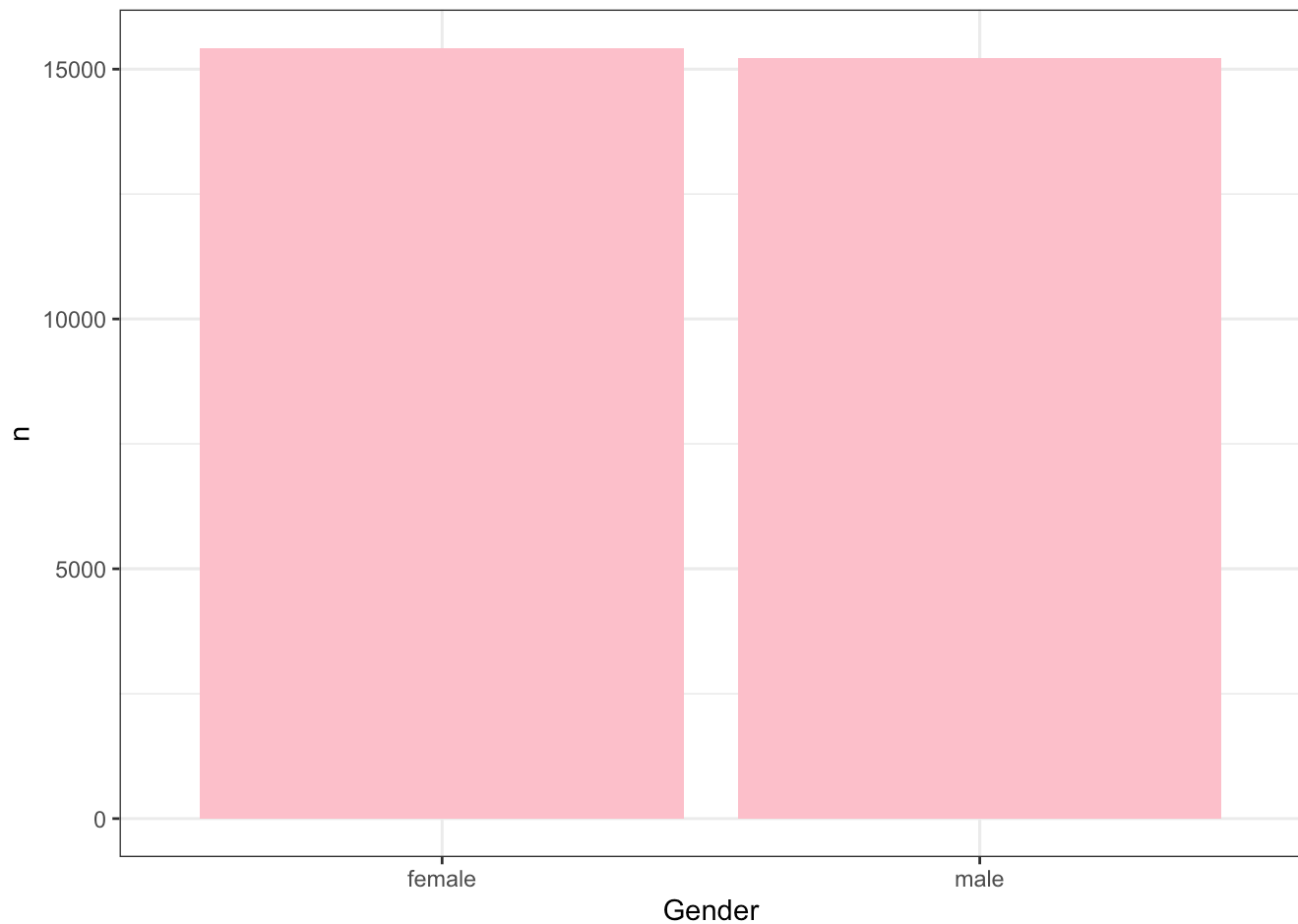
glimpse(education)
```

```
## Rows: 30,641
## Columns: 16
## $ student_ID      <fct> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ...
## $ Gender          <fct> female, female, female, male, male, female, female...
## $ EthnicGroup      <fct> NA, group C, group B, group A, group C, group B, g...
## $ ParentEduc       <fct> bachelor's degree, some college, master's degree, ...
## $ LunchType        <fct> standard, standard, standard, free/reduced, standa...
## $ TestPrep         <fct> none, NA, none, none, none, none, completed, none,...
## $ ParentMaritalStatus <fct> married, married, single, married, married, marrie...
## $ PracticeSport     <fct> regularly, sometimes, sometimes, never, sometimes,...
## $ IsFirstChild      <fct> yes, yes, yes, no, yes, yes, no, yes, no, yes, yes...
## $ NrSiblings        <fct> 3, 0, 4, 1, 0, 1, 1, 1, 3, NA, 1, 1, 1, 1, 2, 0, 0...
## $ TransportMeans    <fct> school_bus, NA, school_bus, NA, school_bus, school...
## $ WklyStudyHours    <fct> < 5, 5 to 10, < 5, 5 to 10, 5 to 10, 5 to 10, 5 to...
## $ MathScore         <dbl> 71, 69, 87, 45, 76, 73, 85, 41, 65, 37, 58, 40, 66...
## $ ReadingScore      <dbl> 71, 90, 93, 56, 78, 84, 93, 43, 64, 59, 54, 52, 82...
## $ WritingScore      <dbl> 74, 88, 91, 42, 75, 79, 89, 39, 68, 50, 52, 43, 74...
## $ School            <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
```

Now all categorical variables are factors and all continuous variables are numeric.

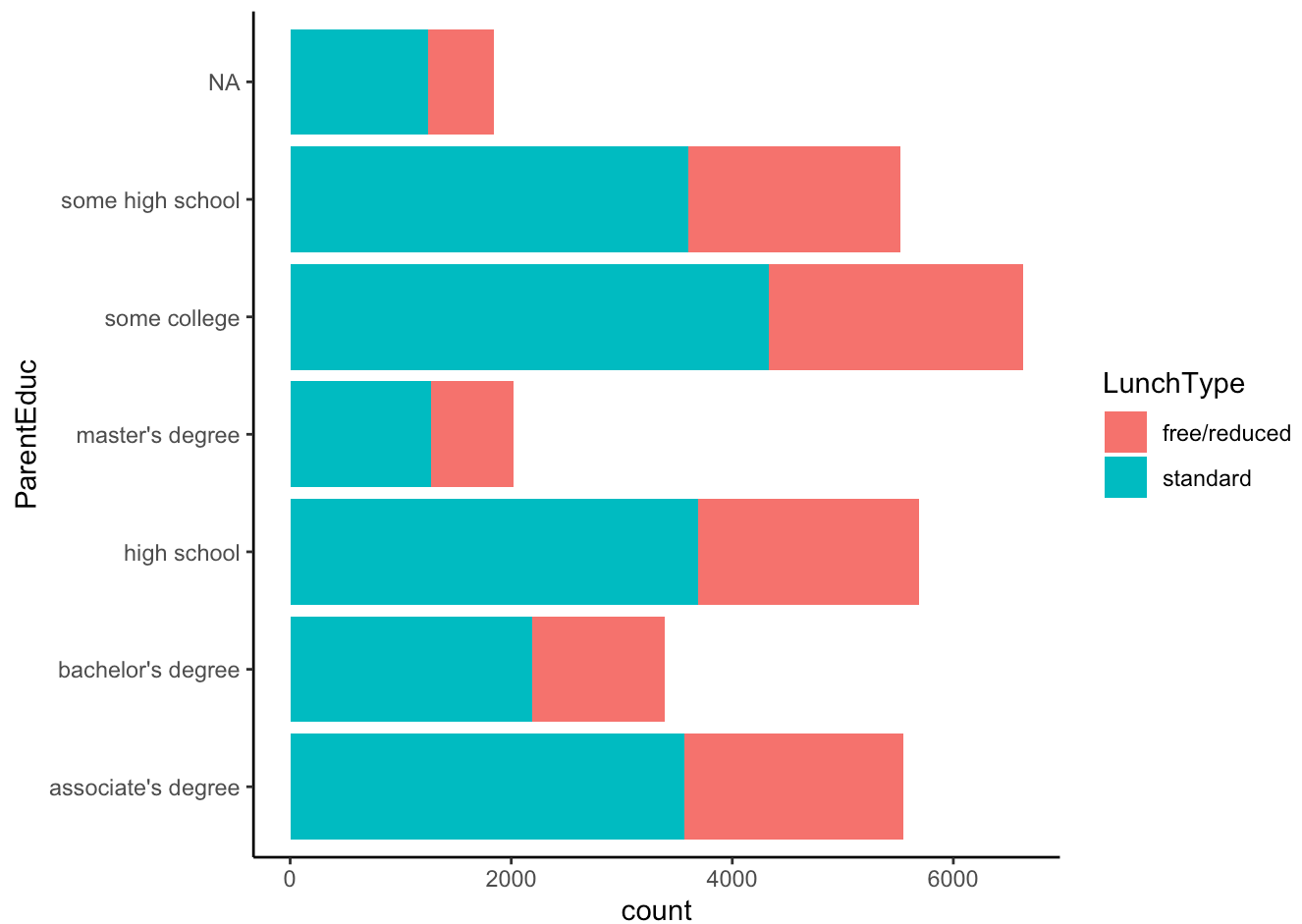
7. Categorical Variables (Simple Bar Chart)

```
education %>% count(Gender) %>%
  ggplot(mapping = aes(x = Gender, y = n)) +
  geom_col(fill = 'pink') +
  theme_bw()
```



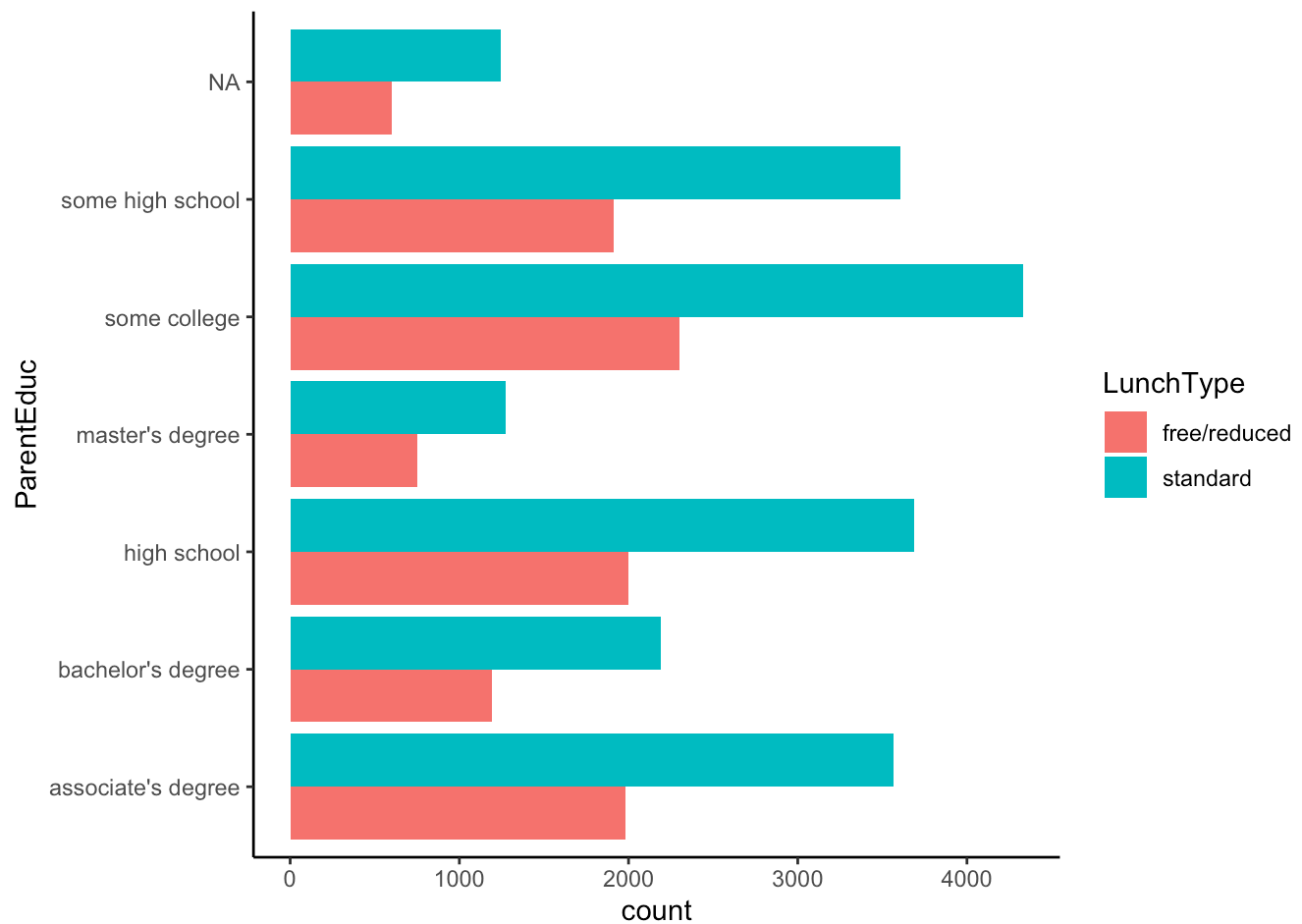
8. Categorical Variables (Stacked Bar Chart)

```
education %>%  
  ggplot(mapping = aes(y = ParentEduc)) +  
  geom_bar(mapping = aes(fill = LunchType)) +  
  theme_classic()
```



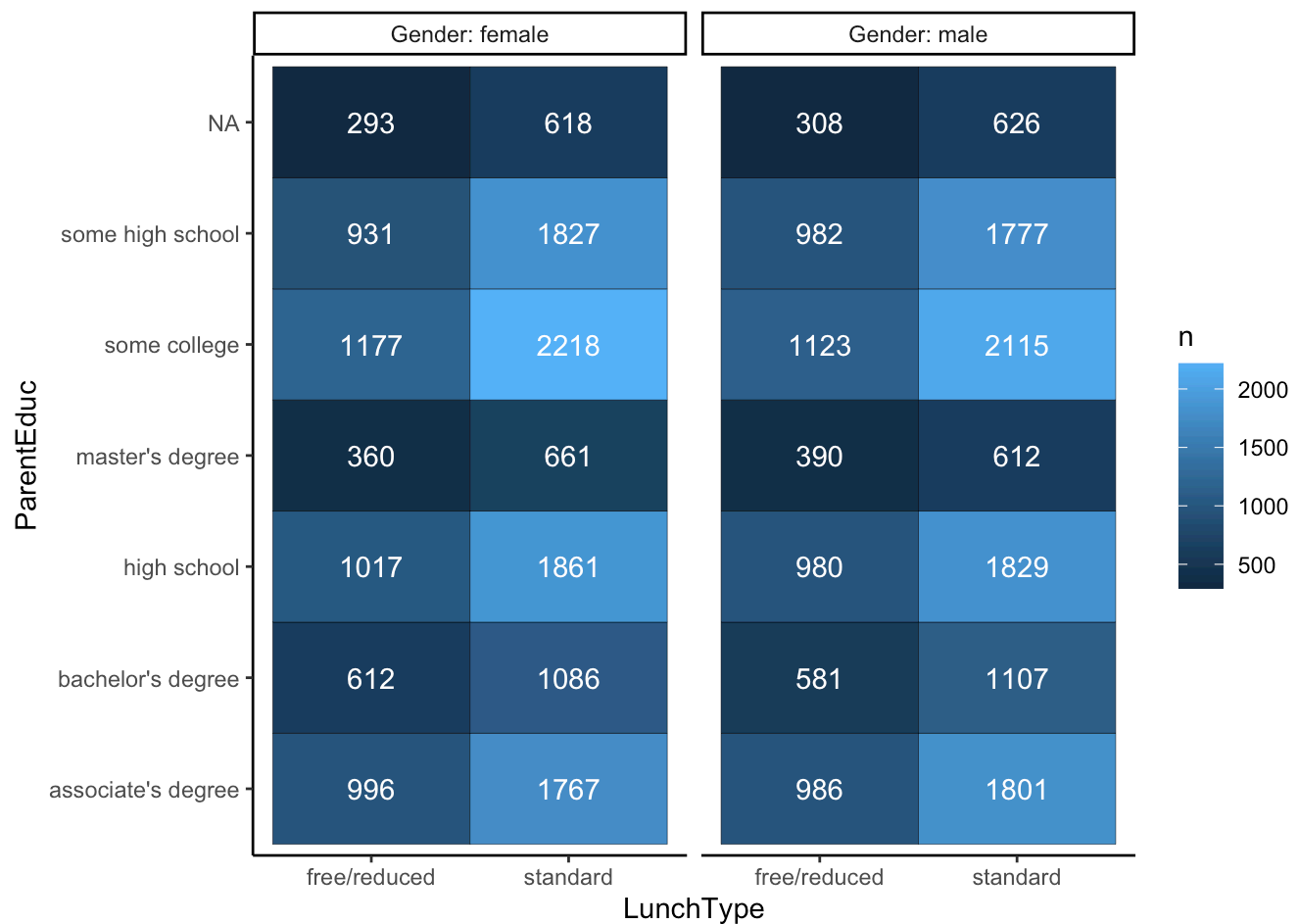
9. Categorical Variables (Dodged Bar Chart)

```
education %>%  
  ggplot(mapping = aes(y = ParentEduc)) +  
  geom_bar(mapping = aes(fill = LunchType), position = 'dodge') +  
  theme_classic()
```



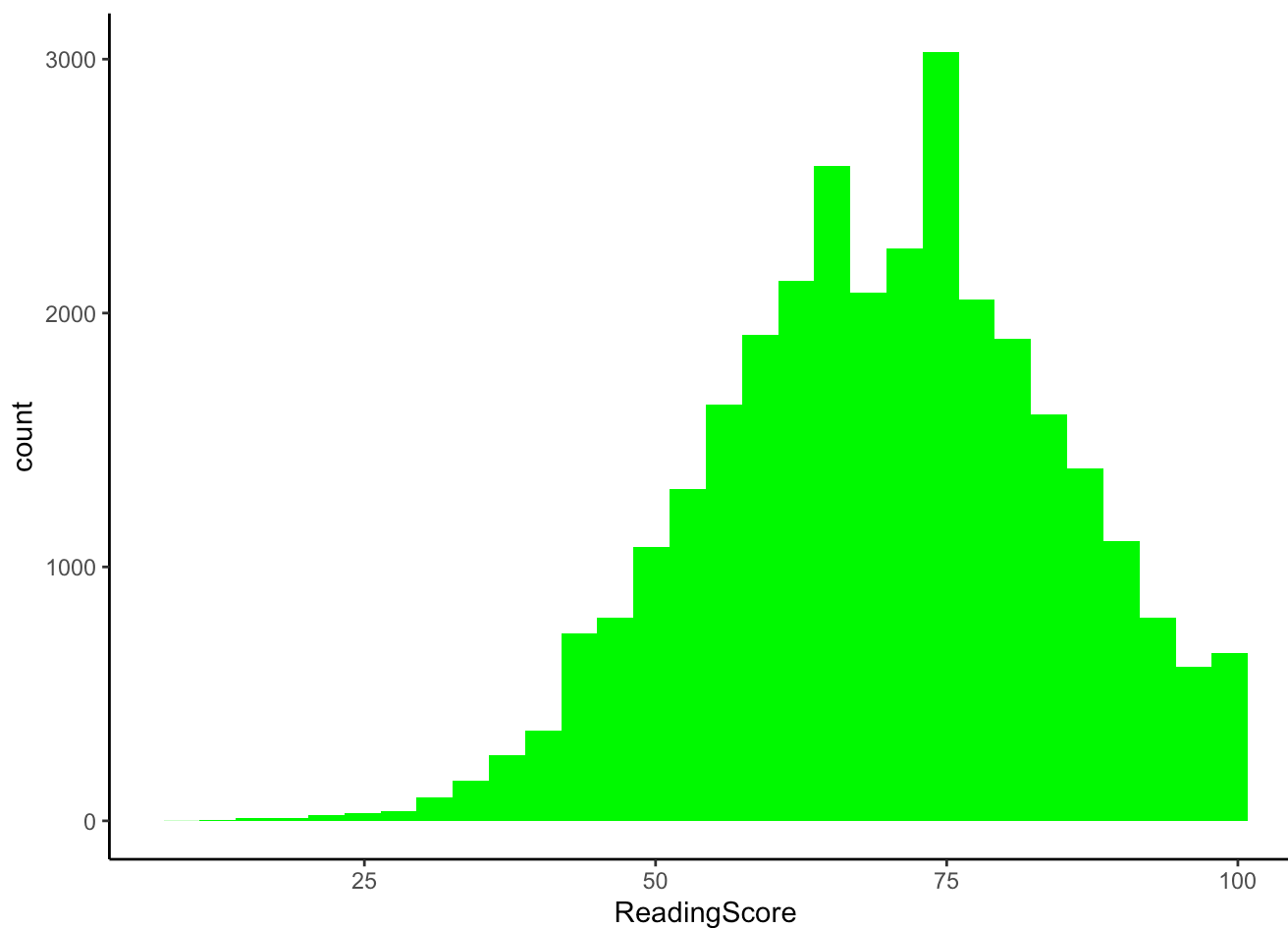
10. Categorical Variables (Heat Map)

```
education %>%
  count(ParentEduc, LunchType, Gender) %>%
  ggplot (mapping = aes(x = LunchType, y = ParentEduc)) +
  geom_tile(mapping = aes(fill = n),
            color = "black") +
  facet_wrap(~ Gender, labeller = 'label_both') +
  geom_text(mapping = aes(label = n), color = 'white') +
  theme_classic()
```



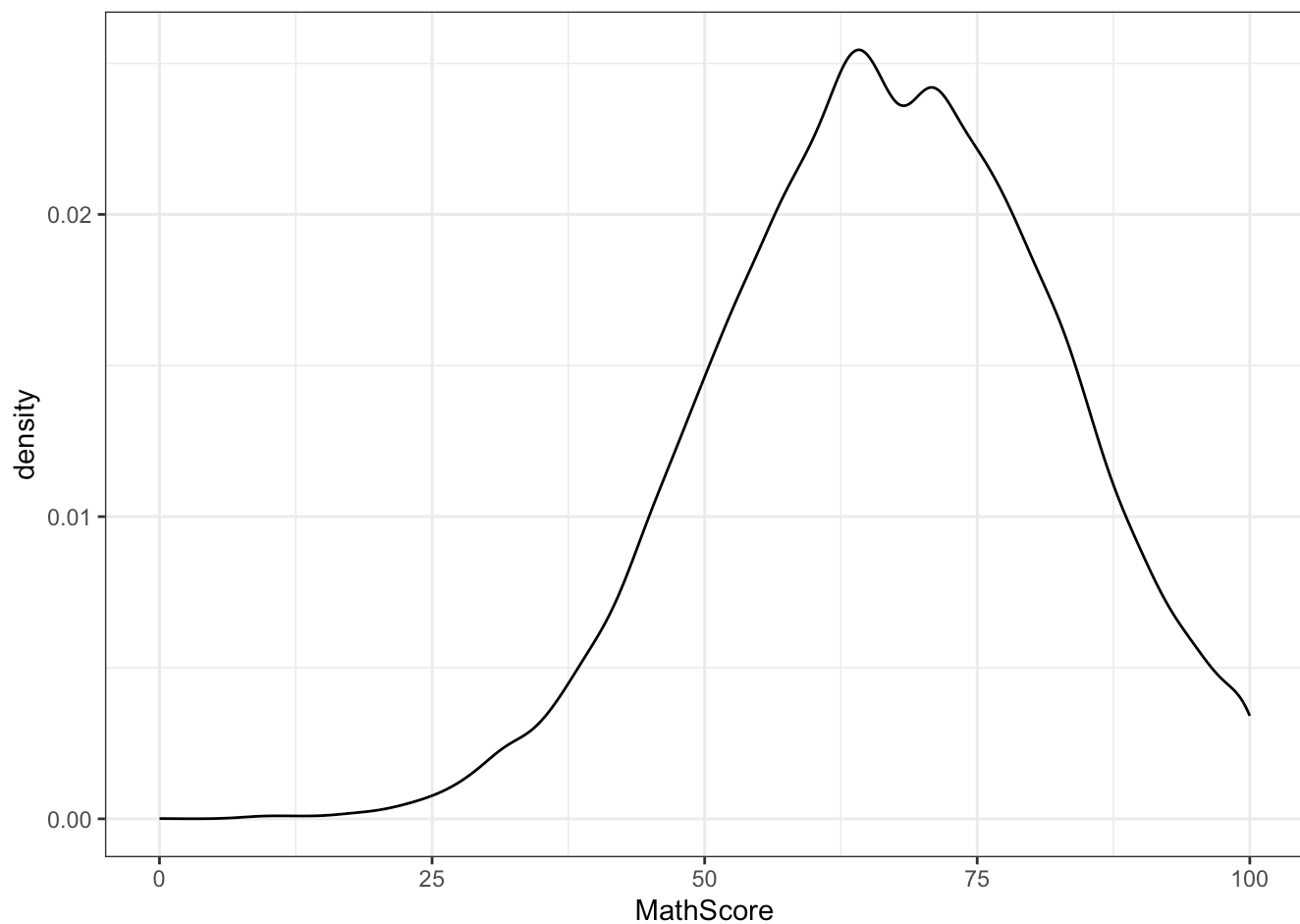
11. Continuous Variables (Histogram)

```
education %>%
  ggplot(mapping = aes(x = ReadingScore)) +
  geom_histogram(bins = 30, fill = 'green') +
  theme_classic()
```

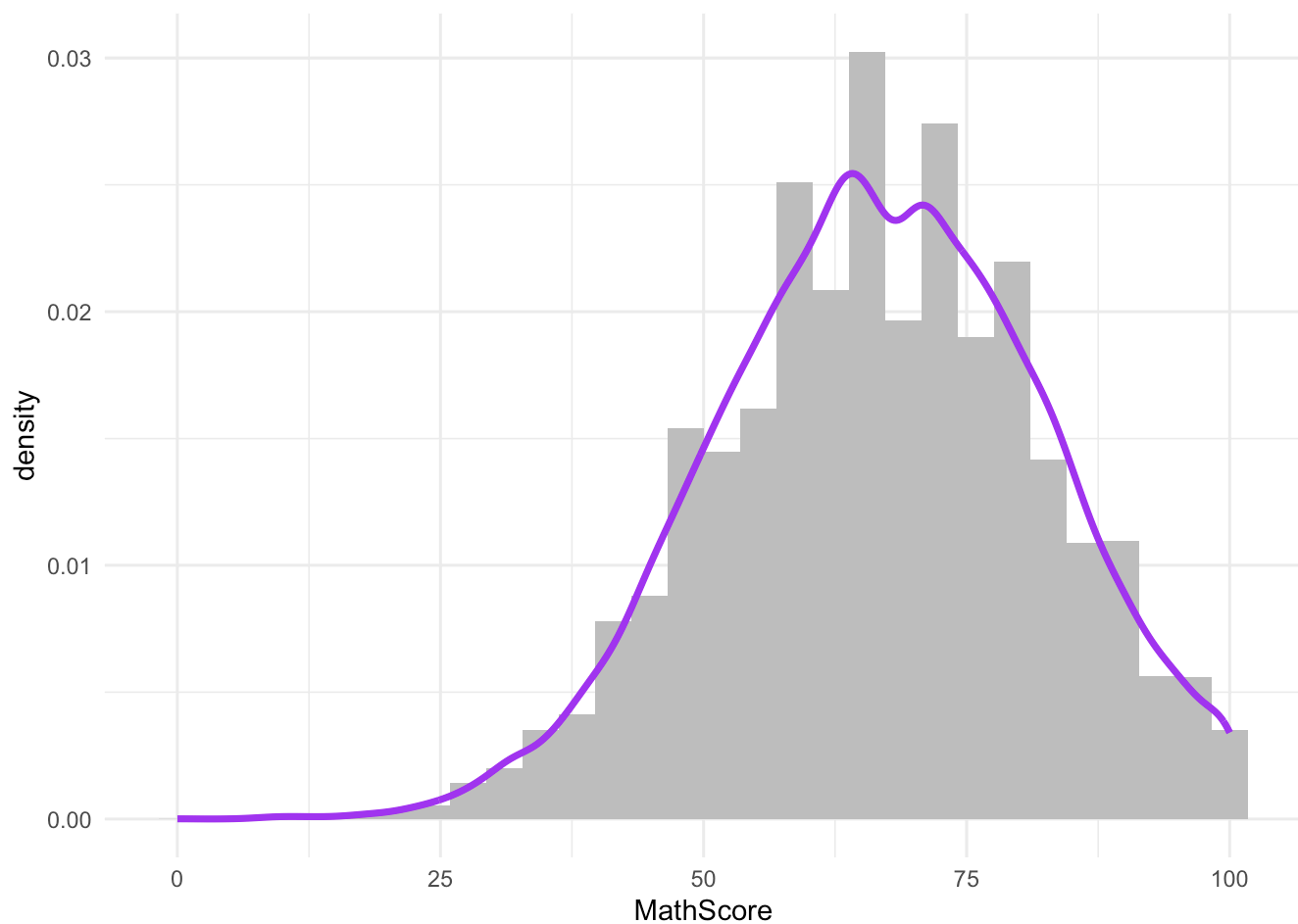
12. Continuous Variables (Density Plot)

```
education %>%  
  ggplot(mapping = aes(x = MathScore)) +  
  geom_density() +  
  theme_bw()
```



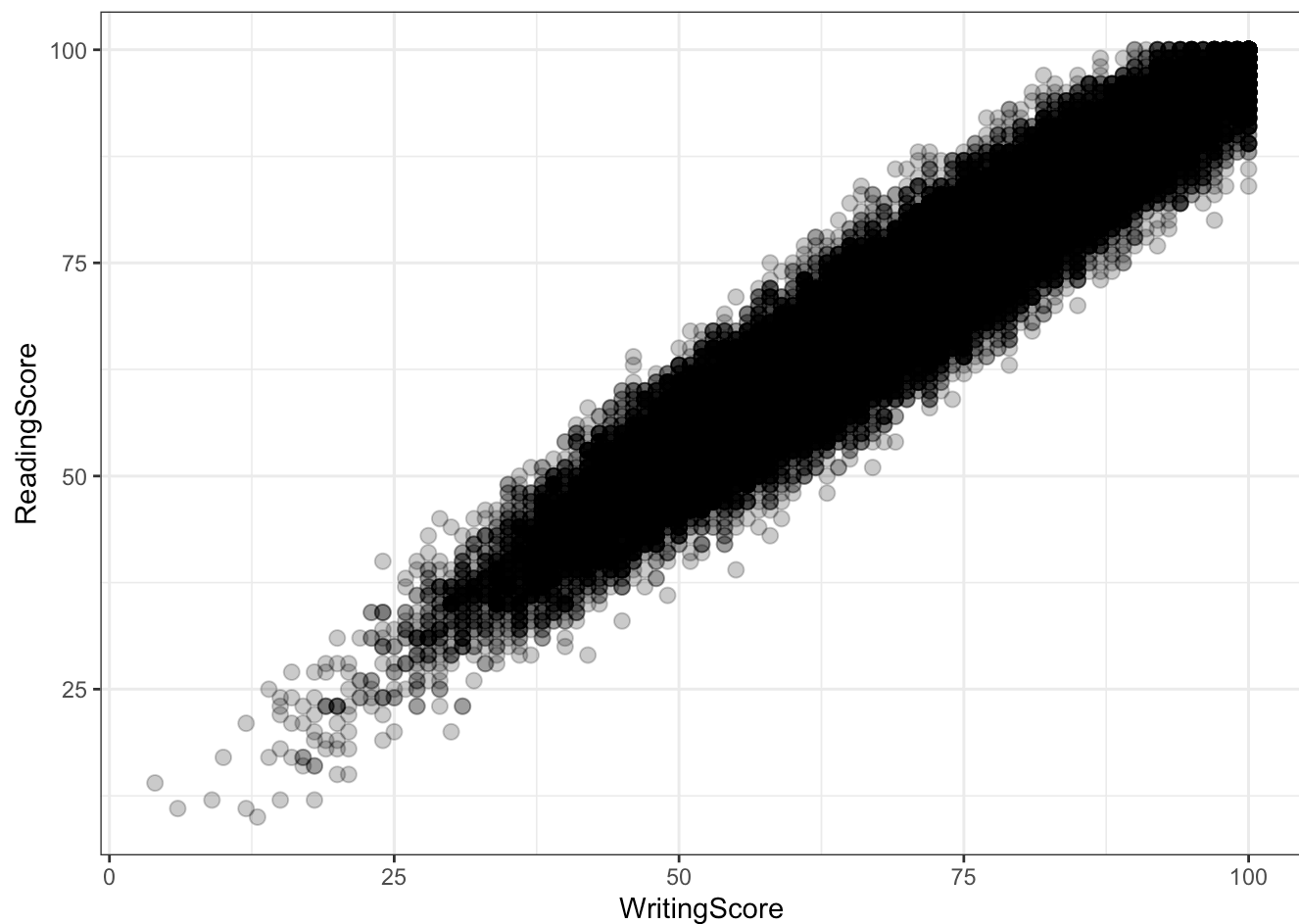
13. Continuous Variables (Histogram + Density Plot)

```
education %>%  
  ggplot(mapping = aes(x = MathScore)) +  
  geom_histogram(bins = 30, fill = 'grey', mapping = aes(y = after_stat(density))) +  
  geom_density(color = 'purple', linewidth = 1.25) +  
  theme_minimal()
```



14. Continuous Variables (Scatterplot)

```
education %>%  
  ggplot(mapping = aes(x = WritingScore, y = ReadingScore)) +  
  geom_point(size = 2.5 , alpha = .25) +  
  theme_bw()
```



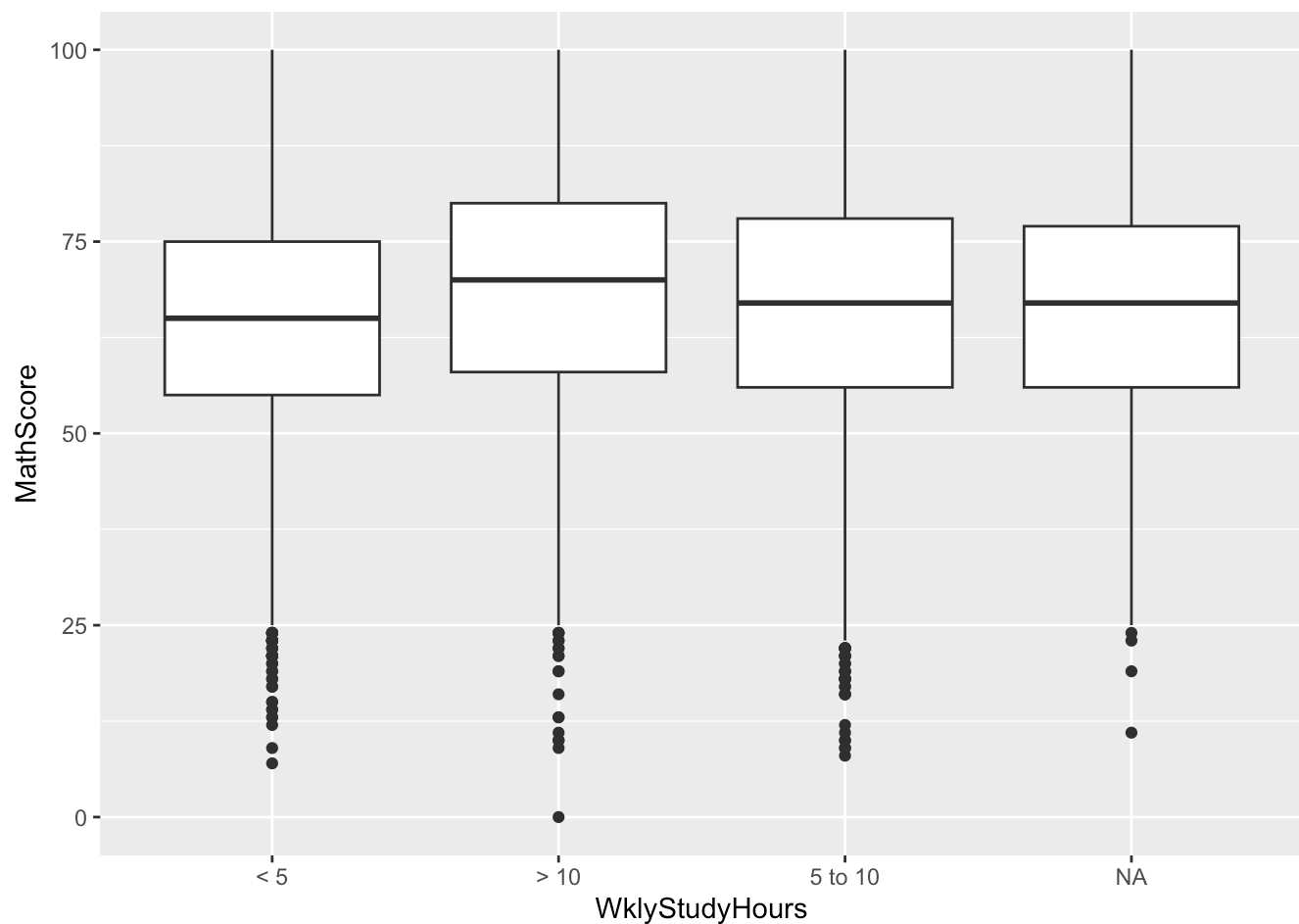
15. Continuous Variables (Correlation Plot)

```
education %>%  
  select(MathScore, ReadingScore, WritingScore) %>%  
  cor() %>%  
  corplot::corrplot(method = 'number', type = 'upper')
```



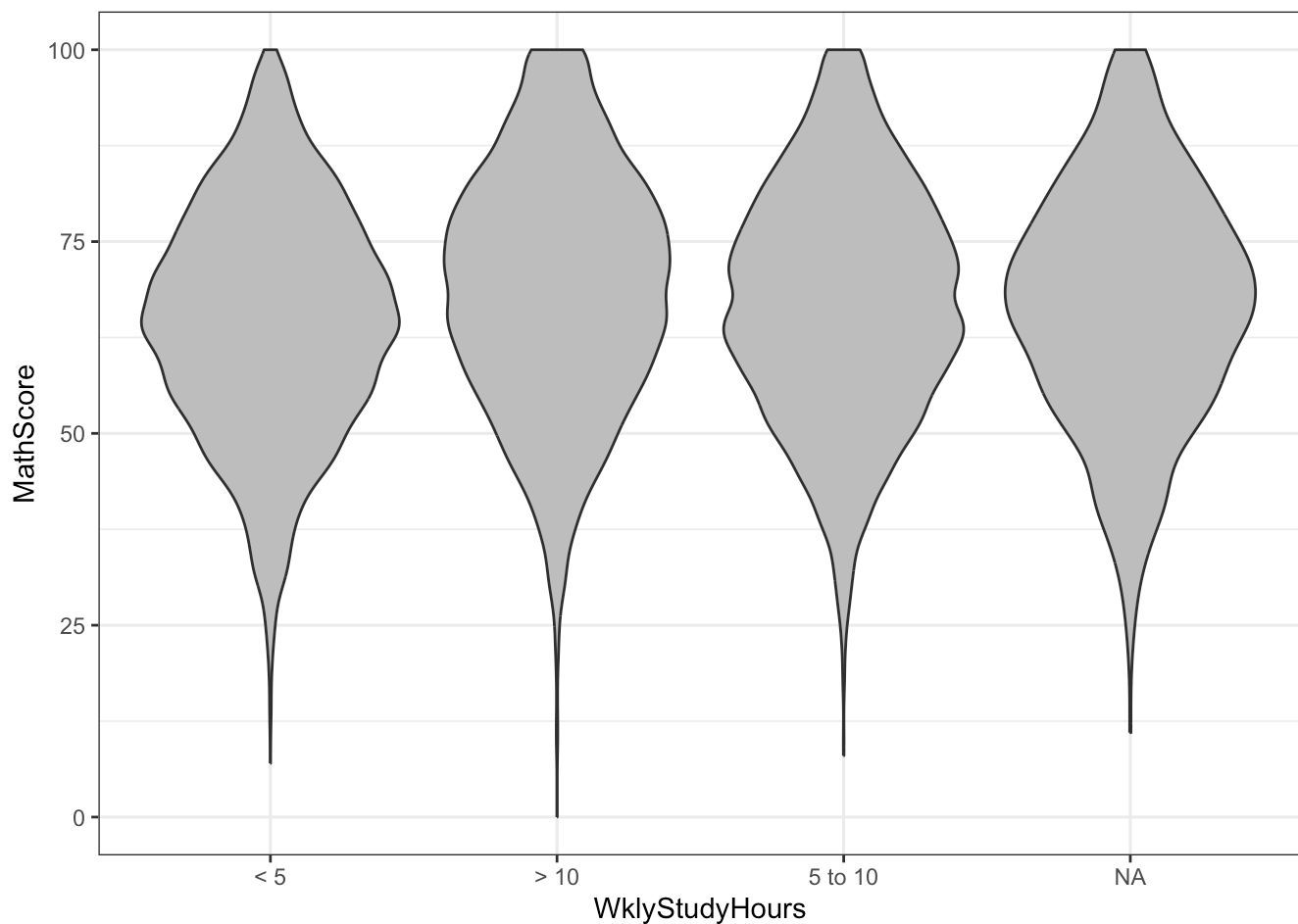
16. Categorical + Continuous Variables (Boxplot)

```
education %>%
  ggplot(mapping = aes(x = WklyStudyHours, y = MathScore)) +
  geom_boxplot()
```



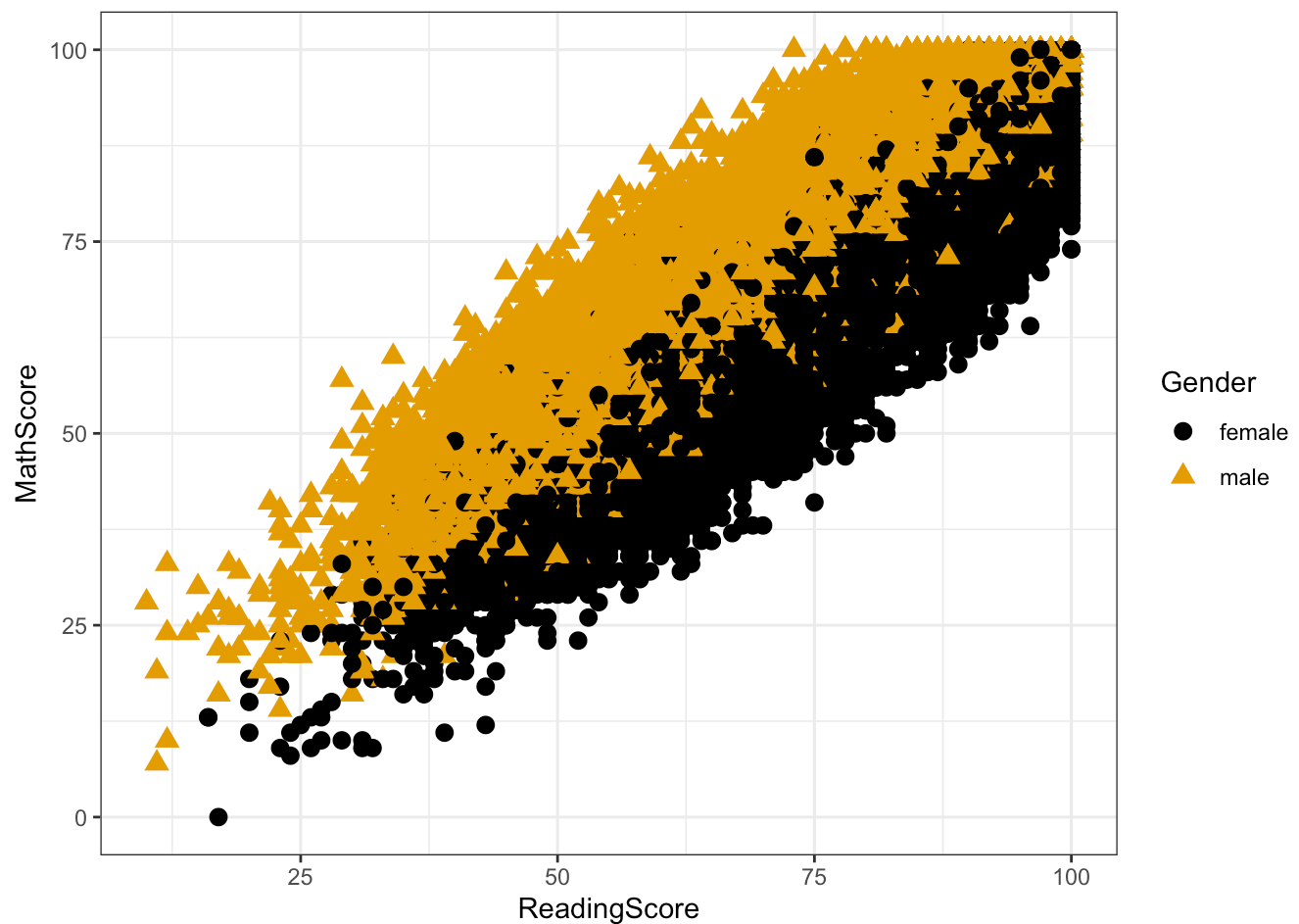
17. Categorical + Continuous Variables (Violin Plot)

```
education %>%  
  ggplot(mapping = aes(x = WklyStudyHours, y = MathScore)) +  
  geom_violin(fill = 'grey') +  
  theme_bw()
```



18. Categorical + Continuous Variables (Scatterplot)

```
education %>%  
  ggplot(mapping = aes(x = ReadingScore, y = MathScore)) +  
  geom_point(mapping = aes(color = Gender, shape = Gender), size = 3) +  
  ggthemes::scale_color_colorblind() +  
  theme_bw()
```



19. Categorical + Continuous Variables (Heat Map)

```
education %>%  
  ggplot(mapping = aes(x = ReadingScore, y = MathScore)) +  
  geom_bin_2d(bins = 11) +  
  facet_wrap(~ ParentEduc) +  
  scale_fill_viridis_c() +  
  theme_bw()
```